

DataCo Supply Chain Performance Report

Delivery reliability, profitability, and delay drivers across 172,765 orders, with a predictive model for late-delivery risk.

Dataset: DataCoSupplyChainDataset.csv · Orders analyzed: 172,765

1. Executive Summary

Across 172,765 completed orders, 54.7% arrived later than their scheduled shipment window — a reliability problem larger than any single region or season. The pattern is structural rather than seasonal: delay rates barely move across months, weekdays, or hours of day, but they move sharply across shipping mode. First Class orders were late 100% of the time and Second Class 79.8% of the time, against 39.8% for Standard Class — evidence that expedited-tier delivery promises are not being met, or are mis-set against actual fulfillment capacity.

The business remained profitable overall (net profit of \$3.81M on \$7.51M gross profit against \$3.71M in losses), and per-order profit does not meaningfully erode as delay length increases — so the immediate margin impact of a late order is small. The real cost of the 54.7% late rate is customer trust and repeat business, not per-order economics. A Random Forest model trained on order attributes predicts late-delivery risk at booking time with 0.83 ROC-AUC, giving a practical lever to intervene before an order ships.

2. Key Performance Indicators

Metric	Value
Total orders	172,765
Late deliveries	94,523 (54.7%)
On-time deliveries	78,242 (45.3%)
90th-percentile delay length	3 days
Gross profit (profitable orders)	\$7,514,814
Gross loss (unprofitable orders)	-\$3,708,393
Net profit	\$3,806,421
Order outcome split	80.7% profit / 18.7% loss / 0.6% break-even

3. Profitability Analysis

Mean profit per order, grouped by delay length in days (negative = shipped early):

Delay (days)	Orders	Mean profit/order	Total profit
-2 (2 days early)	20,873	\$23.36	\$487,596
-1 (1 day early)	20,719	\$21.60	\$447,629

Delay (days)	Orders	Mean profit/order	Total profit
0 (on time)	36,650	\$22.25	\$815,430
+1	53,503	\$22.33	\$1,194,895
+2	27,551	\$21.13	\$582,111
+3	6,772	\$20.03	\$135,653
+4	6,697	\$21.37	\$143,107

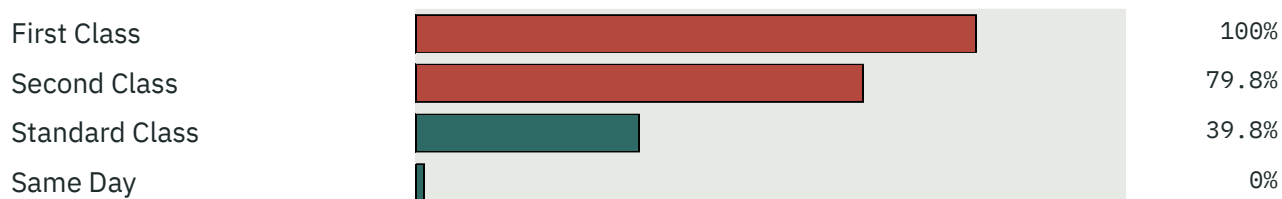
FINDING

Mean profit per order stays within a narrow \$20–23 band regardless of delay length. Late delivery is not materially cannibalizing per-order margin — the \$2.06M in combined profit sitting on delayed orders confirms those orders still monetize normally. The business risk from lateness is reputational and retention-based, not a direct hit to unit economics.

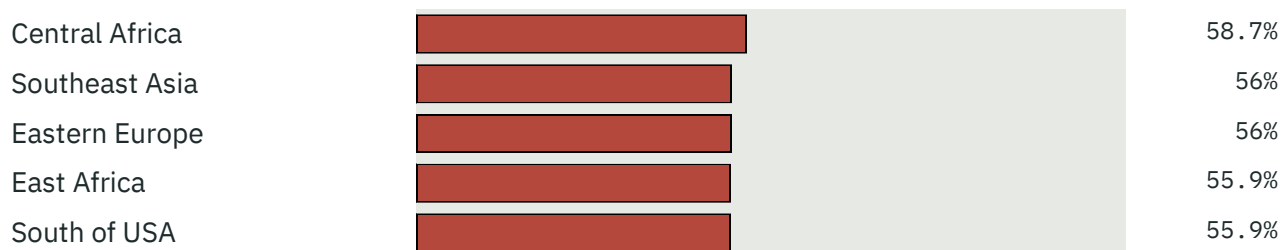
4. Bottleneck Detection

Share of orders delivered late, by segment. Shipping Mode shows the widest spread of any dimension tested.

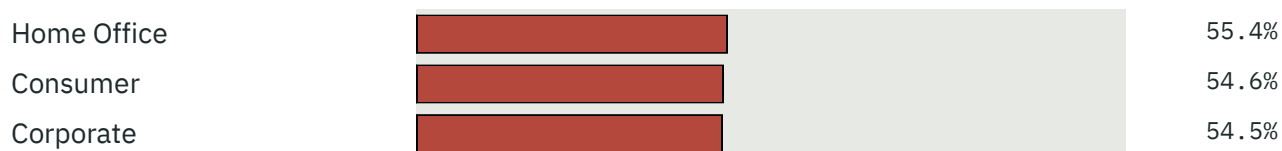
By Shipping Mode



By Order Region (top 5)



By Customer Segment



BOTTLENECK

Shipping Mode is the dominant lever, not geography or customer type — region and segment delay rates cluster tightly in the mid-50% range, while shipping mode ranges from 0% to 100%. This points to the delivery-promise logic (scheduled transit days per mode) as the primary fix target, ahead of any single region's carrier performance.

5. Root Cause Analysis

Deepest drill-down: within Central Africa (the worst-performing region at 58.7% late), which factors push delay rate highest.

Driver	Level	Delay %	Avg delay (d)	Orders
Shipping Mode	First Class	100.0%	1.00	278
Shipping Mode	Second Class	82.8%	2.13	291
Order Status	Payment Review	80.0%	1.20	20
Order Status	Pending	69.1%	0.57	188
Type	Payment	63.6%	0.76	346
Type	Transfer	63.3%	0.61	411
Order Status	Pending Payment	62.6%	0.74	326
Department	Outdoors	61.3%	0.80	80

ROOT CAUSE

The pattern repeats the global one: shipping mode dominates, and order-status values tied to payment processing (Payment Review, Pending, Pending Payment) are the next-strongest driver. This suggests part of the delay is generated upstream of shipping — in how long an order sits before payment clears — rather than in transit itself.

6. Time-Based Delay Patterns

Delay rate by calendar period. Ranges are shown to make clear how flat the pattern is.

Dimension	Highest	Rate	Range across all values
Month	August / September	55.4%	~54.0-55.4% across all 12 months
Day of week	Monday	55.5%	~54.0-55.5% across all 7 days
Hour of day	20:00	57.1%	~53-57% across the day

FINDING

No time dimension shows a meaningful swing - the widest spread (hour of day) is only about 4 points. Delay is not a seasonal or scheduling-capacity problem; it repeats at essentially the same rate every

day, every month, every hour. That reinforces the Section 4-5 conclusion: the cause is structural (shipping-mode promises, payment-processing lag), not staffing or demand peaks.

7. Machine Learning Model Results

Five classifiers trained to predict Late_delivery_risk from order attributes (shipment type, scheduled shipping days, category, segment, department, region, shipping mode, order month/hour), on a class-balanced training set, evaluated on a held-out 20% test set.

Model	Accuracy	Precision	Recall		F1	ROC-AUC
Decision Tree	0.719	0.768	0.730		0.748	0.728
Random Forest	0.738	0.785	0.746		0.765	0.825
KNN	0.697	0.761	0.686		0.722	0.767
Gradient Boosting	0.725	0.893	0.590		0.710	0.778
Logistic Regression	0.697	0.833	0.590		0.690	0.709

Recommended model: Random Forest. Decision Tree edges it out on F1, but a single tree overfits more readily and is less stable on new data. Random Forest's ROC-AUC of 0.825 - clearly ahead of every other model - means its risk scores rank orders by lateness probability more reliably, which matters most for a model meant to flag at-risk orders before they ship, not just classify them after the fact. Gradient Boosting and Logistic Regression trade recall for precision (they catch fewer late orders but are more often right when they flag one) - worth revisiting if the business prefers fewer false alarms over broader coverage.

Top predictive features (Random Forest)

Order hour		25%
Days for shipment (scheduled)		15.3%
Shipping mode		14%
Order month		12.7%
Order region		11.3%
Product category		9.4%

8. Strategic Recommendations

- Investigate payment-processing lag as a delay source. Orders in Payment Review, Pending, and Pending Payment status show elevated delay rates even before shipping - tightening payment turnaround could cut delay independent of logistics.
- Prioritize Central Africa, Southeast Asia, Eastern Europe, and East Africa for carrier and fulfillment-partner review; all sit 4+ points above the segment average.
- Don't over-invest in seasonal or shift staffing fixes. Delay is essentially flat across months, weekdays, and hours - the problem is structural, so capacity-timing changes alone won't move the number.

4. Operationalize the Random Forest risk model to score orders at booking time; route high-risk orders to proactive customer communication or expedited handling.
5. Track delay by customer retention, not margin. Per-order profit is stable regardless of delay, so use repeat-purchase rate and complaint volume - not margin - as the KPI for delivery-reliability investment.

9. Conclusion

DataCo's delivery network is profitable but unreliable: 54.7% of orders miss their scheduled window, and that rate is driven overwhelmingly by shipping-mode promises and payment-processing delays rather than by region, customer type, or time of year. Because per-order profitability barely moves with delay, the immediate financial exposure is limited - but the reputational exposure from delivering late on more than half of all orders, particularly on premium shipping tiers, is significant. The Random Forest late-delivery model (0.825ROC-AUC) gives operations a practical way to flag risk before it happens; pairing that with a fix to how expedited-shipping promises are set is the highest-leverage next step.
